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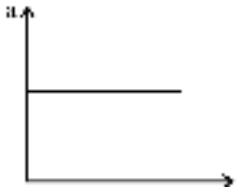
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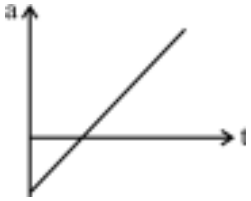
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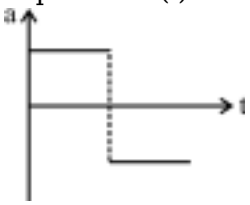
PHYSICS

1) A particle (initially in motion) is subjected to different accelerations separately. These acceleration vary with time as shown. In which of these cases particle will definitely return to its initial position. (i)



(ii)





(iii)

(1) (i) only

(2) (ii) & (iii) only

(3) in all cases

(4) in no case

2) A particle is projected from the ground with an initial speed of v at an angle θ with horizontal. The average velocity of the particle between its point of projection and highest point of trajectory is :-

(1) $\frac{v}{2}\sqrt{1+2\cos^2\theta}$

(2) $\frac{v}{2}\sqrt{1+\cos^2\theta}$

(3) $\frac{v}{2}\sqrt{1+3\cos^2\theta}$

(4) v cos θ

3) The retardation of a moving particle, if the relation between time and position is $t = x^2 + x$ will be-

(1) $2\,(x+1)^{-3}$

(2) $2(2x+1)^{-3}$

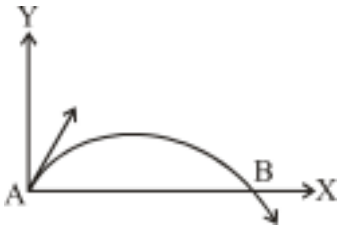
(3) $1/2\,(x+1)^{-3}$

(4) $1/2\,[2x+1]^{-3}$

4) The velocity of a projectile at the initial point A is 2i + 3j m/s then its velocity at point B is :

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(1) 2i + 3j

(2) –2i – 3j

(3) –2i + 3j

(4) 2i – 3j

5) A body A is thrown up vertically from the ground with velocity v_0 and another body B is simultaneously dropped from a height H. They meet at a height $\frac{H}{2}$ if v^0 is equal to.

(1) $\sqrt{2gH}$

(2) $\sqrt{gH}$

(3) $\frac{1}{2}\sqrt{gH}$

(4) $\sqrt{\frac{2g}{H}}$

6) A particle has initial velocity 15 m/s and retardation 2 m/s^2. Distance travelled in 8th second is :-

(1) 2 m

(2) 30 m

(3) 0.5 m

(4) 56 m

7) A cars starts from rest, moves with an acceleration a and then decelerates at a constant rate b for sometime to come to rest. If the total time taken is t, the maximum velocity of car is given by :-

(1) $\frac{abt}{(a+b)}$

(2) $\frac{a^2t}{(a+b)}$

(3) $\frac{at}{(a+b)}$

(4) $\frac{b^2t}{(a+b)}$

8) The equation of motion of a projectile is : $y = 12x - \frac{3}{4}x^2$ Given that g = 10 m/s^2, what is the range of the projectile?

(1) 12 m

(2) 16 m

(3) 30 m

(4) 36 m

9) The engine of a motorcycle can produce a maximum acceleration 5 m/s^2. Its brakes can produce a maximum retardation 10 m/s^2.If motorcyclist start from point A and reach at point B. What is the minimum time in which it can cover if distance between A and B is 1.5 km :-
(Given : that motorcycle comes to rest at B)

(1) 30 sec

(2) 15 sec

(3) 10 sec

(4) 5 sec

10) A cricketer hits a ball with a velocity 25 m/s at 60° above the horizontal. how far above the ground it passes over a fielder 50 m from the bat (assume the ball is struck very close to the ground).

(1) 8.2 m

(2) 9.0 m

(3) 11.6 m

(4) 12.7 m

11) A projectile is thrown in the upward direction making an angle of 60° with the horizontal direction with a velocity of 147 ms^{-1}. Then the time after which its inclination with the horizontal is 45°, is :

(1) 15 s

(2) 10.98 s

(3) 5.49 s

(4) 2.745 s

12) A body falls freely from rest. It covers as much distance in the last second of its motion as covered in the first three seconds. The body has fallen for a time of :

(1) 3 s

(2) 5 s

(3) 7 s

(4) 9 s

13) A particle is projected with velocity v_0 along x-axis. The deceleration on the particle is proportional to the square of the distance from the origin i.e. $a = -x^2$ The distance at which particle stops is

(1) $\sqrt{\frac{3v_0}{2}}$

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(2) $\left(\frac{v_0^2}{3}\right)^{\frac{1}{3}}$

(3) $\left(\frac{2v_0^2}{3}\right)^{\frac{1}{3}}$

(4) $\left(\frac{3v_0^2}{2}\right)^{\frac{1}{3}}$

14) Length of seconds hand is 5 cm. Find the distance travelled by seconds hand in 27 seconds.

(1) $\frac{9\pi}{2}$ cm

(2) $\frac{15\pi}{2}$ cm

(3) $\frac{21\pi}{2}$ cm

(4) $\frac{3\pi}{2}$ cm

15) A point initially at rest moves along x-axis. Its acceleration varies with time as $a = (6t + 5)$ m/s^2. If it starts from origin, the distance covered in 2s is :-

(1) 20 m

(2) 18 m

(3) 16 m

(4) 25 m

16) A ball is projected from the ground at angle θ with the horizontal. After 1 second it is moving at angle 45° with the horizontal and after 2 seconds it is moving horizontally. What is the velocity of projection of the ball :-

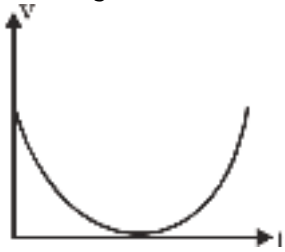
(1) $10\sqrt{3}ms^{-1}$

(2) $20\sqrt{3}ms^{-1}$

(3) $10\sqrt{5}ms^{-1}$

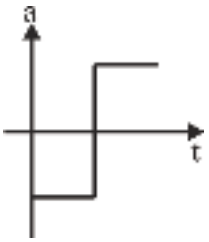
(4) $20\sqrt{2}ms^{-1}$

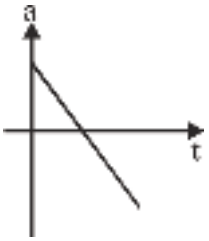
17) The velocity time graph of a moving object is shown in figure. The corresponding acceleration-time graph representing the motion of the object is :-

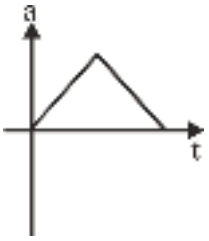


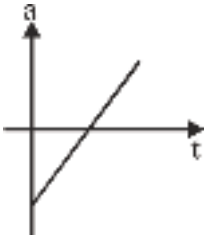
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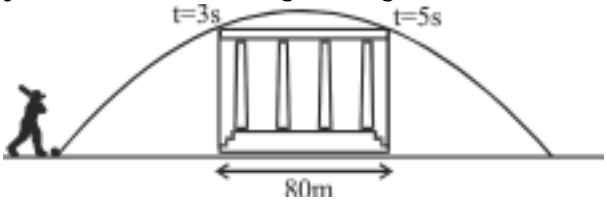
(1) 

(2) 

(3) 

(4) 

18) The six hit by CHRIS GAYLE in IPL just misses a building of length 80 m as shown in figure. The angle of projection with horizontal is :-

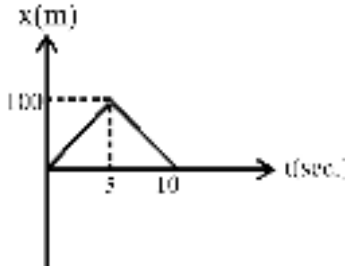


(1) 45°
(2) 30°
(3) 60°
(4) 15°

19) A body goes 30 km south and then 40 km east. What will be the displacement from initial point?

(1) 50 km, 37º South of East
(2) 30 km, 37º South of East
(3) 40 km, 53º South of East
(4) 70 km, 53º South of East

20) Find the displacement of the particle from following position-time graph:



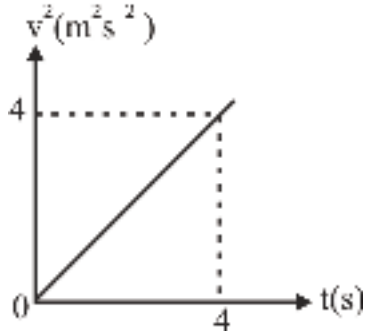
(1) 200 m

(2) 0 m

(3) 10 m

(4) 100 m

21) A particle is moving along a straight line such that square of its velocity varies with time as shown in the figure. What is the acceleration of the particle at t = 4 s ?



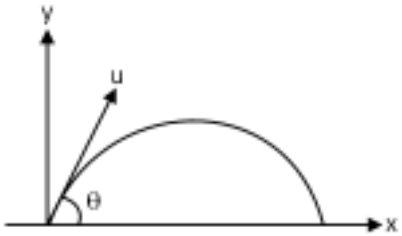
(1) 4 m/s^2

(2) 1/4 m/s^2

(3) 1/2 m/s^2

(4) 0

22) For ground to ground projectile motion with initial velocity u at angle θ from horizontal.



(A)	Change in speed from initial to final point	(i)	u – ucosθ
(B)	Magnitude of change in velocity from initial to final point	(ii)	usinθ
(C)	Reduction in speed from initial to top	(iii)	2usinθ
(D)	Magnitude of change in velocity from initial to top	(iv)	zero

Options :-

(1) (A)–(i), (B)–(ii), (C)–(iii), (D)–(iv)

(2) (A)–(iv), (B)–(ii), (C)–(i), (D)–(iii)

(3) (A)–(iv), (B)–(iii), (C)–(i), (D)–(ii)

(4) (A)–(iv), (B)–(iii), (C)–(ii), (D)–(i)

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23) If a particle is moving along straight line and its x-coordinate as a function of time is given below $x = \frac{2}{3}t^3 - 2t^2 + 2t - 5$ then find the time when direction of velocity changes

(1) At t = 1 sec

(2) At t = 2 sec

(3) At t = 5 sec

(4) Direction of velocity remains unchange

24) Charge density of a solid sphere is given by $\rho = \frac{\rho_0 r}{R}$. Then total charge of sphere will be (where R is radius of sphere) :-

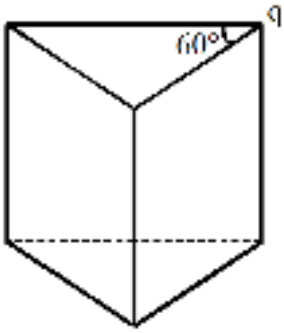
(1) $2\pi R^3\rho_0$

(2) $\pi R^3\rho_0$

(3) $4\pi R^3\rho_0$

(4) $\frac{\pi}{2}R^3\rho_0$

25) A charge q is placed at a vertex of the given geometry of angle 60°. Then, the flux coming out from this geometry will be :



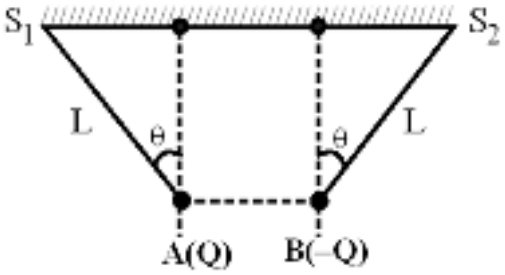
(1) $\frac{q}{6\epsilon_0}$

(2) $\frac{q}{16\epsilon_0}$

(3) $\frac{q}{8\epsilon_0}$

(4) $\frac{q}{12\epsilon_0}$

26) Two balls A and B of same mass (M) and charges Q, –Q are suspended by two strings of same length from two different suspension points S_1 and S_2. If S_1S_2 = 3x and AB = x, then tension in the string will be :



(1) $\frac{Q^2L}{4\pi\varepsilon_0 x^3}$

(2) $\frac{QL}{4\pi\varepsilon_0 x^3}$

(3) $\frac{KQL}{x^2}$

(4) None of these

27) If linear charge density of a wire of length L depends on distance x from one end as $\lambda = \frac{\lambda_0(x)}{2L}$.



Find total charge on wire :-

(1) $\lambda_0 L$

(2) $\frac{\lambda_0 L}{2}$

(3) $\frac{\lambda_0 L}{4}$

(4) None of these

28) Two parallel infinite long straight wire having linear charge densities +λC/m and –2λC/m linear are placed at a distance 2R in free space. What is the electric field mid way between the two line charges ?

(1) $\frac{3\lambda}{2\pi\varepsilon_0 R}$

(2) $\frac{\lambda}{2\pi\varepsilon_0 R}$

(3) $\frac{\lambda}{4\pi\varepsilon_0 R}$

(4) $\frac{3\lambda}{4\pi\varepsilon_0 R}$

29) Two charges of equal mass m are suspended with two strings of equal length □, vertically hanged to a point on a rigid support. At equilibrium, if separation between charges is x; the correct relation between charge q and x is (if angle between strings is very small) :-

(1) $q \propto x^2$

(2) $q \propto x^3$

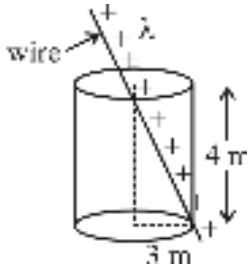
(3) $q \propto x^{3/2}$

(4) $q \propto x^{1/2}$

30) Linear charge density of wire is 8.85 μC/m Radius and height of the cylinder are 3m and 4 m.

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Then find the flux passing through the cylinder :-

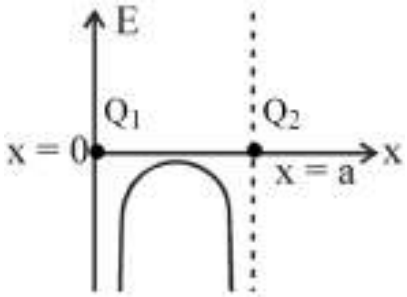
(1) 5×10^6 volt-m

(2) 3×10^6 volt-m

(3) 4×10^6 volt-m

(4) None

31) Two point charges Q_1, Q_2 are fixed at x = 0 and x = a. Assuming that field strength is positive in the direction coinciding with the positive direction of x, then, which following option will be correct ?



(1) Both Q_1 and Q_2 are negative with $|Q_1| > |Q_2|$

(2) Q_1 is positive and Q_2 is negative with $|Q_1| > |Q_2|$

(3) Q_1 is negative and Q_2 is positive with $|Q_1| = |Q_2|$

(4) Both Q_1 and Q_2 are positive with $|Q_1| > |Q_2|$

32) Match the first part of a sentence given in Column-I with its second part in Column-II so that sentence is complete meaningful electrostatically.

Column-I		**Column-II**	
(A)	Like point charges	(i)	Of two types
(B)	Unlike point charges	(ii)	Repel each other
(C)	Charge can be	(iii)	Attract each other
(D)	Charges are	(iv)	Neutralised, if they are equal and opposite

(1) A- i, B- ii, C- iii, D-iv

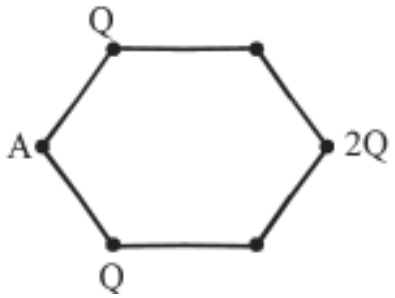
(2) A- ii, B- i, C- iii, D-iv

(3) A- iv, B- i, C- iii, D-ii

(4) A- ii, B- iii, C- iv, D-i

33) Regular hexagon with side length 'a' is shown, find electric field and potential at A.

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(1) $\frac{2KQ}{a^2}, \frac{3kQ}{2a}$

(2) $\frac{3}{2}\frac{KQ}{a^2}, \frac{3kQ}{a}$

(3) $\frac{3}{4}\frac{KQ}{a^2}, \frac{3kQ}{a}$

(4) $\frac{KQ}{a^2}, \frac{3kQ}{2a}$

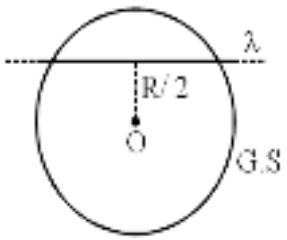
34) Force between two identical charges, placed at distance r apart in vacuum, is F. Now a slab of dielectric constant K = 4 is inserted between these two charges. The thickness of the slab is r/2. The force between the charges will now become :

(1) F/4

(2) F/2

(3) 3F/5

(4) 4F/9



35) Find flux from Gaussian surface (G.S.) (Radius of sphere is R).

(1) $\frac{\sqrt{3}\lambda R}{2\epsilon_0}$

(2) $\frac{\sqrt{3}\lambda R}{\epsilon_0}$

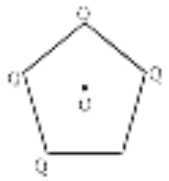
(3) $\frac{\lambda R}{\epsilon_0}$

(4) $\frac{2\lambda R}{\epsilon_0}$

36) In the given diagrams the direction of electric field at point O is given in list-II (O is circumcenter of the given regular polygon). Charge Q is positive. Match the direction of electric field for the given arrangement:-

	List-I		**List-II**

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(P)		1.	
(Q)		2.	
(R)		3.	
(S)		4.	

Code :-

	P	Q	R	S
(1)	2	1	4	3
(2)	3	1	2	4
(3)	3	2	1	4
(4)	1	4	3	2

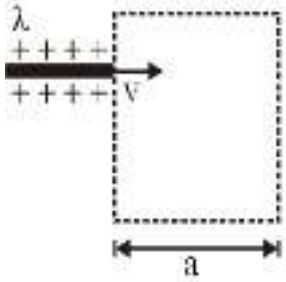
(1) 1

(2) 2

(3) 3

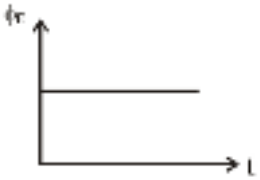
(4) 4

37) Figure shows an imaginary cube of side a. A uniformly charged rod of length $\frac{a}{2}$ moves towards right at a constant speed v. At t = 0, the right end of the rod just touches the left face of the cube.

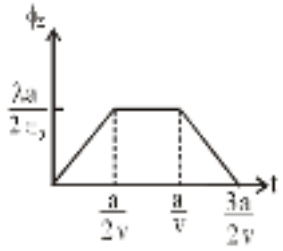


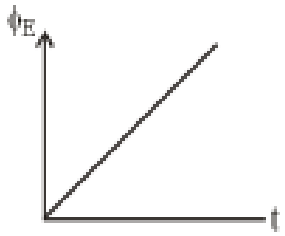
Plot a graph between electric flux passing through the cube versus time :-

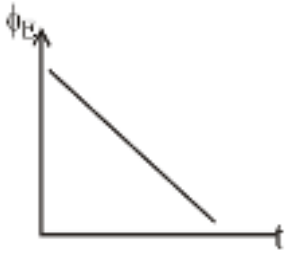
(1)



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(2) 

(3) 

(4) 

38) Find total number of electric field lines originate from +2.15 μC charge.

(1) 2.5×10^5

(2) 5×10^5

(3) 8×10^5

(4) None of these

39) Two identical small spheres carry charge of Q_1 and Q_2 with $Q_1 >> Q_2$. The charges are d distance apart. The force they exert on one another is F_1. The sphere are made to touch one another and then separated to distance 'd' apart. The force they exert on one another now is F_2, then $\frac{F_1}{F_2}$ is :

(1) $\frac{4Q_1}{Q_2}$

(2) $\frac{Q_1}{4Q_2}$

(3) $\frac{4Q_2}{Q_1}$

(4) $\frac{Q_1}{4Q_2}$

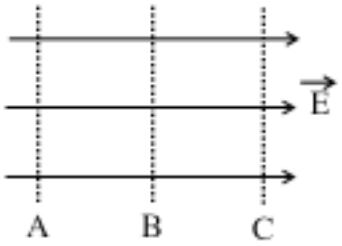
40) Electric field on the axis of ring is :

(1) $\frac{kqx}{(r^2+x^2)^{\frac{3}{2}}}$

(2) $\frac{kqx}{\sqrt{r^2+x^2}}$

(3) $\frac{kq}{(r^2 + x^2)^{\frac{3}{2}}}$

(4) $\frac{kq}{r^2}$



41) Find relation between E_A, E_B & E_C.

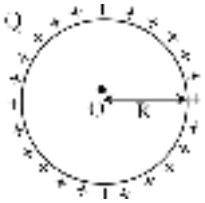
(1) $E_A = E_B > E_C$

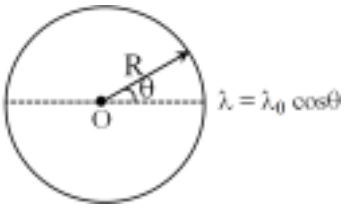
(2) $E_A = E_B = E_C$

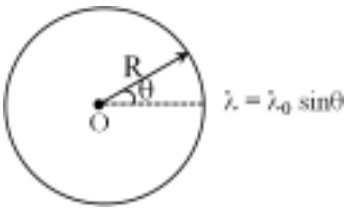
(3) $E_A > E_B > E_C$

(4) $E_A < E_C$

42) In which of the following figure, electric field at point O is non zero ?

(1) 

(2) 

(3) 

(4) (2) and (3) both are correct

43) There exists a uniform electric field $\vec{E} = \left(\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}\right)$ N/C in a region. Calculate the flux of electric field through a circle of radius R. (equation of circle ⇒ $x^2 + y^2 = 4$)

(1) α β γ

(2) 4πγ

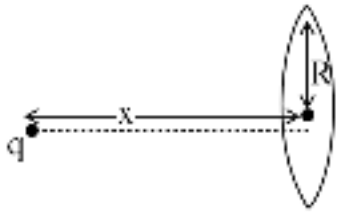
(3) 2πα

(4) 2πβ

44) A point charge q is placed at origin. Find the flux of electric field, produced due to this charge,

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through a circle of radius R with the help of fig. (R >>>x)

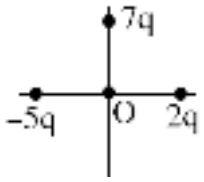
(1) $\frac{q}{\epsilon_0}$

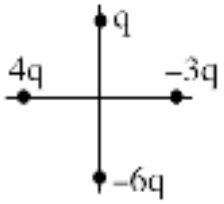
(2) ER^2

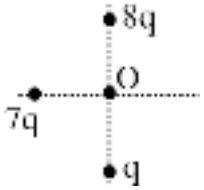
(3) $\frac{q}{2\epsilon_0}$

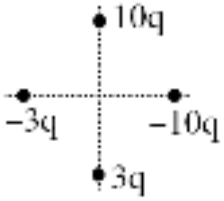
(4) $E \cdot 4\pi R^2$

45) The figure shows four situations in which charged particles are at equal distances from the origin. E_1, E_2, E_3 and E_4 are the magnitude of fields at the origin in each case. Select the correct alternative.

(I)


(II)


(III)


(IV)


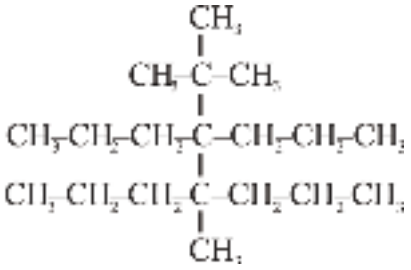
(1) $E_1 > E_2 > E_3 > E_4$

(2) $E_1 < E_2 < E_3 < E_4$

(3) $E_1 = E_2 = E_3 = E_4$

(4) $E_1 > E_2 < E_3 < E_4$

CHEMISTRY



1) Total number of carbon atom present in parent chain is :-

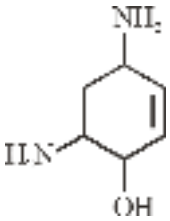
(1) 5

(2) 6

(3) 7

(4) None

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2) IUPAC name of compound is :- 

(1) 4, 6-diamino cyclohex-2-en-1-ol
(2) 2, 4-diamino cyclohex-4-en-1-ol
(3) 4-hydroxy cyclohex-3-en-1, 5-diamine
(4) 1, 5-diamino-4-hydroxy cyclohex-2-ene

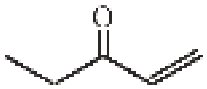
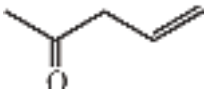
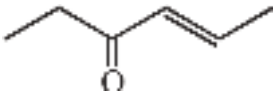
3) Which of the following have only 1° Hydrogen atom :-

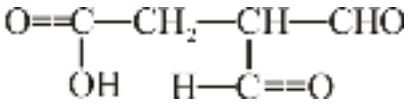
(1) Cyclopropane
(2) 2-methyl propene
(3) Isobutane
(4) 2-methyl butene

4) Which one of the following represent correct order of functional groups according to priority table ?

(1) $-SO_3H > -CN > -CONH_2 > -OH$
(2) $-COX > -OH > -CHO > CN$
(3) $-COOR > -CN > -CHO > -NH_2$
(4) $-SO_3H > -COOH > CHO > -CN$

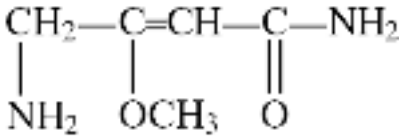
5) Pent-4-en-2-one is the correct IUPAC name for:-

(1) 
(2) 
(3) 
(4) 

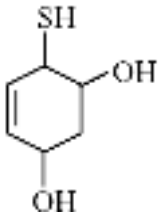
6) The correct IUPAC name of the following compound  is :-

(1) 3,3-Diformyl propanoic acid
(2) 3-Formyl-4-oxobutanoic acid
(3) 3,3-Dioxo propanoic acid
(4) 3,3-Dicarbaldehyde propanoic acid

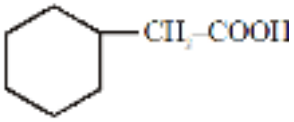
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7) The IUPAC name of the compound is :- 

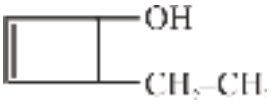
(1) 4–Amino-2–methoxy–1–amino–2–butene
(2) 4–Amino–3–methoxy–2–butenamide
(3) 2–Methoxy–1,4–diamino–2–butenal
(4) 1–Amino–2–methoxy–3–amino propene

8) Following compound is named as :- 

(1) 6–Mercaptocyclohex–4–ene–1,3–diol
(2) 1–Mercaptocyclohex–2–ene–4,6–diol
(3) 1–Mercaptocyclohex–5–ene–2,4–diol
(4) 4–Mercaptocyclohex–2–ene–1,5–diol

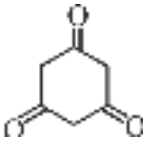
9) How many carbon present in P.C.C of given structure. 

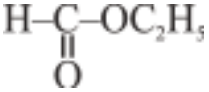
(1) 3
(2) 2
(3) 6
(4) 8

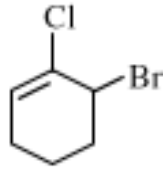
10) IUPAC name of the given compound is :- 

(1) 2–Ethyl cyclobut 3–en–1–ol
(2) 4–Ethyl cyclobut–2–en–1–ol
(3) 3–Ethyl cyclobut–1–en–4–ol
(4) 4–Ethyl cyclobut–1–en–3–ol

11) Which is correct IUPAC name :-

(1)  cyclohexane-1,3,5-trione

(2)  Ethyl methanoate

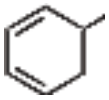
(3) 

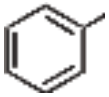
6-Bromo-1-chloro cyclohexene

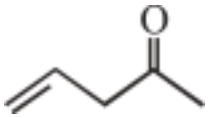
(4) All of these

12) Which of the following molecules have only sp^2 Hybridised carbon atoms ?

(1) 

(2) 

(3) 

(4) 

13) Select the incorrect statement :-

(1) s-block metal oxides are basic in nature except BeO which is amphoteric.
(2) P-block non-metal oxides are acidic in nature except NO, N_2O, CO.
(3) In d-block all elements are metals
(4) In p-block all elements are metals

14)

Which of the following is amphoteric oxide?

Mn_2O_7, CrO_3, Cr_2O_3, CrO, V_2O_5, V_2O_4 :-
(1) V_2O_5, Cr_2O_3
(2) Mn_2O_7, CrO_3
(3) CrO, V_2O_5
(4) V_2O_5, V_2O_4

15)

Which of the following statements is incorrect ?

(1) With the decrease the electronegativity in a group the metallic character increases
(2) Smaller ionization energy value indicates that electropositive character is more
(3) Electronegativity depend upon stability of e^- configuration
(4) Successive ionization energies always increase for an element

16)

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Which of the following is wrong?

(1) Zero EA : He, N, Be

(2) IE : $O^+ > S^+$

(3) IE : $O^- > S^-$

(4) EN : B > Tl > In > Ga > Al

17) **Assertion :** F atom has less electron affinity than Cl atom.
Reason : Additional electrons are repelled more strongly by 3p electrons in Cl atom than by 2p electrons in F atom.

(1) Both Assertion and Reason are correct and reason is correct for the Assertion.

(2) Both Assertion and Reason are correct but Reason is not correct for Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Assertion is incorrect but Reason is correct.

18) **Assertion :** Electron affinity of oxygen is less than that of fluorine but greater than that of nitrogen.
Reason : Order of Ionisation energy is the following:
N > O > F

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

19) Which of the following species is neither hypervalent nor hypovalent?

(1) ClO_4^-

(2) BCl_3

(3) SO_4^{2-}

(4) CO_3^{2-}

20) Which is not possible :-

(1) ICl_3

(2) I_3^-

(3) CCl_6^{-2}

(4) BeF_4^{-2}

21)

IF_3 is formed in :-

(1) 1st excited state of iodine

(2) 2nd excited state of iodine

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(3) Ground state of iodine

(4) All wrong

22) Correct statement is :-

(1) Chlorine can show covalency 3, 5 & 7.

(2) Maximum covalency of 'N' can be 4.

(3) SF_4 exist but OF_4 does not exist.

(4) All are correct

23) In which of the following molecular ions number of □.p are same in central atom and terminal atom.

(1) $I_3^{\ominus}$

(2) $IC\ell_2^{\ominus}$

(3) XeF_2

(4) All of these

24)

Identify correct match

	Type of overlapping	**Bond formed**	**Internuclear axis**
(1)	$p_x + p_x$	π	X
(2)	$p_x + d_{xz}$	π	X
(3)	$p_y + d_{xy}$	π	X
(4)	$s + p_z$	π	X

(1) 1

(2) 2

(3) 3

(4) 4

25)

Element	**IP_1**	**IP_2**	**IP_3**	**IP_4**
A	5.390	75.6	122.4	–
B	8.2	25.1	37.9	259.2
C	9.3	18.2	217.6	217.6

Select correct option (If A, B & C are II period element) :

(1) $A_{(g)} + e \rightarrow \overset{\ominus}{A}_{(g)}$ = +ve

(2) $Z_B > Z_C$ Z = Atomic number

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(3) Oxide of B = Amphoteric

(4) $A_{(g)} \rightarrow A^{\ominus}_{(aq)}$ is non-spontaneous

26) **Assertion (A) :** $Al(OH)_3$ soluble in NaOH but $Fe(OH)_3$ is insoluble in NaOH.
Reason (R) : NaOH forms water soluble complex $Na[A1(OH)_4]$.

(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.

(2) Both Assertion and Reason are true and the Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is false.

(4) Both Assertion and Reason are false.

27) Which is/are correct about electro-negativity order of the following elements ?

(1) A1 > Ga

(2) C > N

(3) C < Br

(4) Sr > Ca

28)

The hydroxide of which metal ion is not soluble in excess of sodium hydroxide solution :

(1) $A1^{+3}$

(2) Cr^{3+}

(3) Be^{+2}

(4) Cu^{2+}

29) Which of the following oxide is different than other ?

(1) SO_2

(2) Cl_2O_7

(3) PbO_2

(4) Mn_2O_7

30) Incorrect statement about valency is :

(1) It increases from IA to IVA with respect of hydrogen.

(2) It increase from IA to VIIA with respect to Hydrogen.

(3) Maximum covalency of second period elements is four.

(4) Third period element can extend their covalency (more than 4).

31) Incorrect statement about nitrogen is :

(1) General covalency = 3

(2) Maximum covalency = 4

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(3) Maximum oxidation state = +5

(4) In N_2O_4 and N_2O_5, covalency = 3, 4

32) The combination of orbital that produce non-bonding molecular orbital is (inter nuclear axis is Z-axis).

(1) $P_z + P_z$

(2) $p_z + d_{yz}$

(3) $P_y + d_{yz}$

(4) $d_{xy} + d_{xy}$

33) Which pair of element can form multiple bond with itself and oxygen ?

(1) F, N

(2) N, C1

(3) N, P

(4) N, C

34) Which is correct entry in blanks ?

	Orbits	**Internuclear axis**	**Bond**
i.	d_{xy}, d_{xy}	A	π
ii.	p_x, p_x	Y	B
iii.	p_x, C	Z	π
iv	d_{xy}, p_y	D	π

(1) A – all axis

(2) B = σ or π bond

(3) C = p_x

(4) D = y axis

35) Select the incorrect orders of IP ?

(1) $O^- < S^-$

(2) $F^{\ominus} < Cl^{-}$

(3) $P^{\ominus} < S^{\ominus}$

(4) $O^{\ominus} > Br^{\ominus}$

36) For a first order reaction
$3A(g) \rightarrow 2B(g) + 3C(g) + D(g)$
if initial pressure of A is P_0 and after time 't' the pressure of gaseous mix is P_t the Rate Constant is

(1) $K = \frac{1}{t} \ell n \frac{P_0}{2P_0 - P_t}$

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(2) $K = \frac{1}{t}\ell n\frac{2P_0}{2P_0 - P_t}$

(3) $K = \frac{1}{t}\ell n\frac{P_0}{P_0 - P_t}$

(4) None of these

37) For the reaction A + 2B + 3C → D
following data was obtained. Using the data calculate the order of reaction

Experiment	[A]	[B]	[C]	Rate of reaction (M/sec)
1	0.6 M	1.2 M	0.3 M	0.5
2	0.6 M	2.4 M	0.3 M	1
3	1.2 M	2.4 M	0.3 M	2
4	0.6 M	1.2 M	0.9 M	4.5

(1) 3

(2) 2

(3) 4

(4) 5

38)

	Column-I		**Column-II**
P.	Zero order reaction	1.	$t_{1/2} \propto \frac{1}{[A]_0}$
Q.	First order reaction	2.	$t_{100\%} = [A]_0/k$
R.	Second order reaction	3.	Involves at least two reactants
S.	Pseudo unimolecular reaction	4.	$[A]_t = [A]_0 e^{-kt}$

Code :

	P	Q	R	S
(1)	2	1	4	2
(2)	2	4	1	3
(3)	2	1	3	4
(4)	3	2	1	4

(1) 1

(2) 2

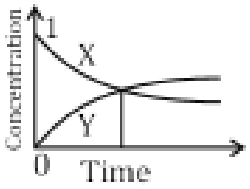
(3) 3

(4) 4

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39) The accompanying figure depicts the change in concentration of species X and Y for the reaction $X \rightarrow 3Y$ as a function of time the point of intersection of the two curves represents.



(1) $t_{1/2}$

(2) $t_{3/4}$

(3) $t_{1/4}$

(4) Data insufficient to predict

40) The rate of first order reaction is 1.5×10^{-2} mol L^{-1} min^{-1} at 0.5 M concentration of the reactant. The half life of the reaction is

(1) 7.53 min

(2) 0.383 min

(3) 23.1 min

(4) 8.73 min

41) Which is/are **correct statements** :
(a) On Increasing **temperature rate of reaction** increases for endothermic and decreases for exothermic reaction.
(b) On Increasing temperature rate of reaction increase due to increase in number of effective collisions.
(c) Collision that can cross energy barrier as well as orientation barrier are effective collision.
(d) For complex molecules rate = $PZ_{AB}\, e^{-Ea/RT}$ acc. to collision theory

(1) a,b

(2) b,c

(3) b,c,d

(4) a,b,c,d

42) For two unimolecular reaction
$k_1 = 10^{12}\, e^{-4000/T}$
$k_2 = 10^{13}\, e^{\frac{-5500}{T}}$
Then at which temperature both reaction will have same ROR.

(1) 432.32

(2) 651.32 k

(3) 215.71k

(4) 912.52k

43) Which is/are statement are correct.
(a) On comparing two reaction, higher E_a reaction will be slower compare to lower E_a reaction (same

arrhenius factor)
(b) Reaction having lower E_a values are more sensitive toward temperature.
(c) Value of rate constant depend on activation energy, catalyst and temperature.
(d) Slope of $\log_{10} K$ and $\frac{1}{T}$ is equal to $\frac{-E_a}{R}$

(1) a,b,c,d
(2) b,c,d
(3) a,c
(4) b,d

44) For a reaction A $\xrightarrow[E_a]{K}$ D
Steps (i) A → B E_a = 40 KJ/mole
$A = 10^6$ sec^{-1}
(ii) B → C E_a = 60 KJ/mole
$A = 10^2$ sec^{-1}
(iii) C → D E_a = 30 KJ/mole
$A = 10^4$ sec^{-1}
If $K_{overall} = \left(\frac{K_1}{K_2}\right)^{3/2} K_3^2$
Then value of $A_{overall}$ and $(E_a)_{overall}$ are :

(1) 10^4 sec^{-1} and 40 KJ/mole
(2) 10^{14} sec^{-1} and 30 KJ/mole
(3) 10^{10} sec^{-1} and 10 KJ/mole
(4) 10^6 sec^{-1} and 60 KJ/mole

45) Which is/are statement correct.
(a) If $E_a = 0$ then rate constant is same at 25°C and at 45°C.
(b) For a reaction if frequency factor is 1.5×10^{14} sec^{-1} and rate constant at 25°C is 5.3×10^{-9} sec^{-1}. Then rate constant at $T = \infty$ is 1.5×10^{14} sec^{-1}
(c) If reaction obey $\log K = 6 - \frac{2000}{T}$ then E_a = 9.212 k cal

(1) a,b only
(2) a only
(3) a,b,c
(4) b,c only

BIOLOGY

1) A child born with a tail like structure is an example of

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(1) Mutation
(2) Variation
(3) Atavism
(4) Acquired character

2) The site of fertilization in human females is usually the:

(1) Ovary
(2) Uterus
(3) Cervix
(4) Ampullary region of fallopian tube

3) Hormone responsible for milk ejection after delivery:

(1) Estrogen
(2) Progesterone
(3) Oxytocin
(4) Prolactin

4) Vasectomy is a method of contraception involving cutting of:

(1) Urethra
(2) Vas deferens
(3) Epididymis
(4) Testis

5) Which of the following organisms connects reptiles and birds in terms of features ?

(1) Icthyosaurus
(2) Peripatus
(3) Archaeopteryx
(4) Neopilina

6) One ovum is released during:

(1) Menstruation
(2) Ovulation
(3) Fertilization
(4) Implantation

7) Part of sperm that contains enzymes to penetrate ovum:

(1) Tail
(2) Head
(3) Acrosome
(4) Midpiece

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8) Which hormone is secreted only during pregnancy?

(1) LH
(2) hCG
(3) FSH
(4) Estrogen

9) The correct sequence is:

(1) Zygote → Morula → Blastocyst → Implantation
(2) Zygote → Blastocyst → Morula → Implantation
(3) Zygote → Implantation → Morula → Blastocyst
(4) Morula → Zygote → Blastocyst → Implantation

10) Spermatogenesis starts at:

(1) Childhood
(2) Puberty
(3) Birth
(4) Menopause

11) Identify a permanent method of contraception:

(1) Condom
(2) Diaphragm
(3) Tubectomy
(4) Oral pill

12) The organ that stores and matures sperm:

(1) Testis
(2) Vas deferens
(3) Epididymis
(4) Urethra

13) Placenta performs all EXCEPT:

(1) Gas exchange
(2) Nutrient transport
(3) WBC production
(4) Hormone secretion

14) Which organ is showing evolutionary loss of function ?

(1) Pancreas
(2) Intestine

(3) Lungs
(4) Wisdom teeth

15) Marsupials of Australia and placental mammals elsewhere develop similar adaptations independently. This represents

(1) Atavism
(2) Parallel evolution
(3) Divergent evolution
(4) Adaptive radiation

16) What is the function of FSH in males?

(1) Induce ovulation
(2) Stimulates testosterone release
(3) Stimulates spermatogenesis
(4) Maintain corpus luteum

17) Which hormone initiates milk production?

(1) Oxytocin
(2) Prolactin
(3) Estrogen
(4) Relaxin

18) Which is NOT a part of the male accessory glands?

(1) Prostate gland
(2) Bulbourethral gland
(3) Seminal vesicle
(4) Testis

19) Duration of human gestation is approximately:

(1) 240 days
(2) 10 months
(3) 9 months
(4) 300 days

20) Which layer of uterus is shed during menstruation?

(1) Myometrium
(2) Perimetrium
(3) Endometrium
(4) Chorion

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21) Which hormone causes ovulation?

(1) FSH
(2) LH
(3) Estrogen
(4) Progesterone

22) Darwin finches evolved different types of beak on the Galapagos Islands to adapt to different food sources. This is an example of

(1) Adaptive radiation
(2) Genetic drift
(3) Atavism
(4) Fossil history

23) Structure that nourishes the embryo in early stages:

(1) Uterus
(2) Endometrium
(3) Placenta
(4) Fallopian tube

24) Match the column:

Column A		**Column B**	
a	Vas deferens	1.	Enzyme release
b	Acrosome	2.	Sperm transport
c	LH	3.	Ovulation trigger
d	Amnion	4.	Embryo protection

(1) a-2, b-1, c-3, d-4
(2) a-1 b-2 c-3 d-4
(3) a-4 b-3 c-2 d-1
(4) a-3 b-4 c-1 d-2

25) Identify the incorrect pair:

(1) Saheli - Centchroman
(2) Condom - Barrier
(3) Copper-T - Hormonal method
(4) Tubectomy - Surgical method

26) Which of these contraceptives prevents ovulation?

(1) Copper-T

(2) Condom
(3) Oral pills
(4) Diaphragm

27) Which of the following is NOT part of male reproductive system?

(1) Epididymis
(2) Uterus
(3) Vas deferens
(4) Prostate

28) Hormone released from corpus luteum:

(1) LH
(2) FSH
(3) Progesterone
(4) HCG

29) Which of the following is an example of biochemical evidences of evolution

(1) Similar bone structure in vertebrates
(2) embryonic similarities
(3) Similarity in DNA and protein
(4) Fossil

30) Abiogenesis was experimentally proven by

(1) Aristotle
(2) Urey-miller
(3) Oparin-Haldene
(4) Muller

31) Barrier method of contraception includes:

(1) Saheli
(2) Copper-T
(3) Condom
(4) Injectable

32) Semen is alkaline due to:

(1) HCO_3^- secretion
(2) Seminal plasma
(3) Cowper's gland
(4) All of these

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33) In males, testosterone is secreted by:

(1) Sertoli cells
(2) Leydig cells
(3) Spermatogonia
(4) Spermatids

34) Menstrual phase is triggered by:

(1) High progesterone
(2) Low FSH
(3) Decrease in estrogen and progesterone
(4) High LH

35) Implantation occurs about how many days after fertilization?

(1) 1-2
(2) 3-4
(3) 6-7
(4) 10-12

36) If no fertilization occurs, which hormone decreases?

(1) LH
(2) Progesterone
(3) Estrogen
(4) FSH

37) Identify the labeled structure "A" in male reproductive system . What is its function?



(1) Sperm production
(2) Spermatid nourishment
(3) Sperm storage and maturation
(4) Testosterone production

38) Number of vestigeal organs given below are
Nictitating membrane
Appendix
Wisdom teeth

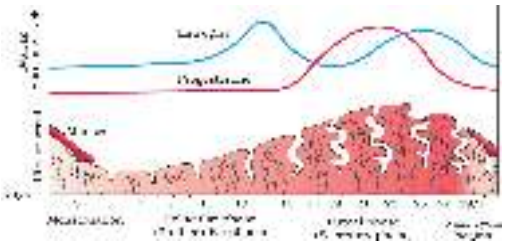
Pinna
Nails

(1) 3
(2) 2
(3) 1
(4) 5

39) Which is incorrect?

(1) Copper-T prevents implantation
(2) Condom prevents STDs
(3) Saheli causes ovulation
(4) Prolactin helps in milk production

40) Identify from graph when ovulation occurs.



(1) Day 1
(2) Day 7
(3) Day 14
(4) Day 28

41) Male reproductive system includes paired testes, accessory ducts, and glands. Name the gland that contributes fructose and prostaglandins to semen.

(1) Prostate
(2) Bulbourethral
(3) Seminal vesicle
(4) Epididymis

42) **Assertion (A) :** Placenta is an endocrine structure.
Reason (R) : It secretes hCG, progesterone, and estrogen

(1) Both A and R are true and R explains A
(2) Both A and R are true but R doesn't explain A
(3) A is true, R is false
(4) A is false, R is true

43) Arrange the following :
1. Molecules
2. Prebiotic soup
3. Protobionts

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4. Chemo heterotrophs
5. Autotrophs

(1) 1-2-3-4-5
(2) 2-1-3-4-5
(3) 1-3-2-4-5
(4) 3-1-2-4-5

44) Match the Column

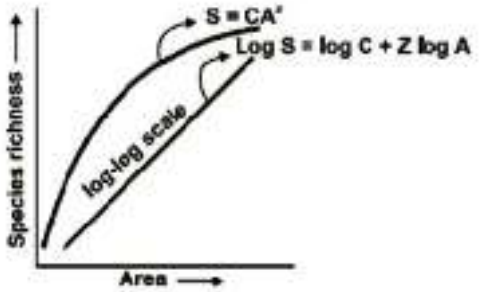
a	Saheli	i	Oral hormonal
b	Condom	ii	IUCD
c	Copper-T	iii	Barrier
d	Tubectomy	iv	Surgical

(1) a-i,b-ii,c-iii,d-iv
(2) a-ii,b-i,c-iv,d-iii
(3) a-iii,b-i,c-ii,d-iv
(4) a-i,b-iii,c-ii,d-iv

45) **Statement 1**: Spermatogenesis requires both FSH and testosterone.
Statement 2: Sertoli cells are stimulated by LH.

(1) Both correct
(2) Both incorrect
(3) Only 1 correct
(4) Only 2 correct

46) Which of the following is correct ?



(1) S = exponential growth, C = Y intercept. Z = Unknown factor, A = area
(2) S = species richness, C = Y intercept, Z = regression coefficient, A = area
(3) S = Logistic growth, C = carrying capacity Z = Environmental resistance, A = area
(4) S = Semi arid species, C = Y intercept. Z = regression coefficient, A = area

47) Which of the following is considered a hot- spot of biodiversity in India?

(1) Indo- Gangetic Plain

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(2) Eastern Ghats

(3) Aravalli Hills

(4) Western Ghats

48)

Match the column-I with column-II and III :-

	I		**II**		**III**
(1)	Kaziranga National park	(a)	White tiger	(i)	Uttarakhand
(2)	Gir National park	(b)	Great indian bustard	(ii)	Rajasthan
(3)	Sundarbans	(c)	Asiatic lion	(iii)	Assam
(4)	Desert National park	(d)	Tiger	(iv)	Gujarat
(5)	Nanda Devi	(e)	Rhinoceros	(v)	West Bengal

(1) 1–b–iii, 2–c–iv, 3–d–v, 4–a–ii, 5–e–i

(2) 1–d–v, 2–b–ii, 3–e–iii, 4–c–iv, 5–a–i

(3) 1–e–iii, 2–c–iv, 3–d–v, 4–b–ii, 5–a–i

(4) 1–a–i, 2–d–v, 3–c–iv, 4–b–ii, 5–e–iii

49) Which one of the following pairs of geographical areas show maximum biodiversity in our country :-

(1) Sunderbans and Rann of Kutch

(2) Eastern Ghats and West Bengal

(3) Eastern Himalaya and Western Ghats

(4) Kerala and Punjab

50) Which one of the following is known as "Lungs of the planet" ?

(1) Amazon rain forest

(2) Western ghats

(3) Indo Burma region

(4) Eastern ghats

51) All of the following are included in 'Ex-situ conservation' except

(1) Wildlife safari parks

(2) Sacred groves

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(3) Botanical gardens
(4) Seed banks

52) Which of the following is not concerned with loss of biodiversity ?

(1) Alien species invasion
(2) Habitat loss and fragmentation
(3) Deforestation
(4) Ex-situ conservation

53) The organization which publishes the Red List of species is :-

(1) ICFRE
(2) IUCN
(3) UNEP
(4) WWF

54) Tiger is not a resident in which one of the following national park ?

(1) Jim Corbett
(2) Ranthambore
(3) Sunderbans
(4) Gir

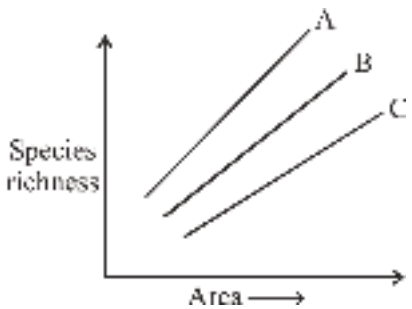
55) Which one of the following represents a renewable source of energy ?

(1) Petroleum
(2) Coal
(3) Nuclear fuel
(4) Trees

56) Which of the following is true :-

(1) 32% of amphibian species face the threat of extinction.
(2) Lichen have higher biodiversity than algae and Fungi together.
(3) Ganga basin have higher biodiversity than Amazon rain forest.
(4) A stable community should show huge variation in productivity from year to year.

57)



Which of the following is/are correct regarding the above mentioned graph?

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(A) A show more species diversity than B and C
(B) Value of Z is less for C in comparison to A and B
(C) Value of Z is maximum for A
(D) C show maximum species diversity

(1) A, B, C and D
(2) A, B and C
(3) B, C and D
(4) A and C

58) What is the most important cause driving animals and plants to extinction ?

(1) Habitat loss & fragmentation
(2) Over - exploilation
(3) Alien species invasion
(4) Co-extinction

59) **Assertion:** The Amazon rain forest is so huge that it is called the lungs of the planet.
Reason: The Amazon rain forest is estimated to produce 20% of the total oxygen in the earth's atmosphere.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

60) **Assertion:** India is one of the 12 mega diversity countries of the world.
Reason: India shares of the global species diversity is an impressive 8.1 percent.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

61) **Assertion:** Bioprospecting is exploring molecular, genetic and species level diversity for the product of economic importance.
Reason: More than 25 percent of the drugs currently sold in the market world wide are derived from plants.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

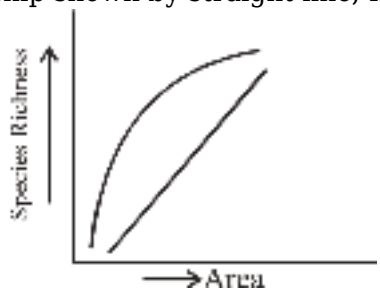
62) **Assertion:** Nature's biological library is burning even before we catalogue the titles of all the books stocked there.
Reason: A large fraction of species diversity is facing the threat of becoming extinct even before we

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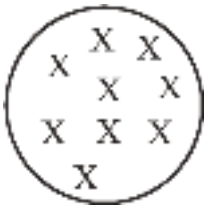
discover them.

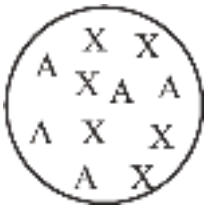
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

63) In the below graph relation between species richness and area for a wide variety of taxa is shown. On a logarithmic scale, the relationship shown by straight line, In which area slope of line to be much steeper ? Here Area (A) = 1 km^2



(1) 

(2) 

(3) 

(4) 

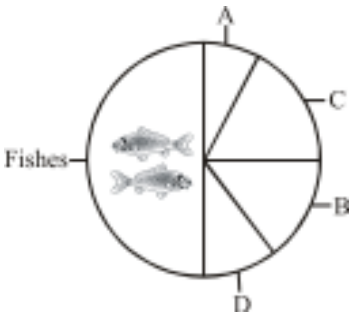
64) How many genetically different strains of rice and varieties of mango are present in India ?

(1) 10,000 and 50,000 respectively
(2) 10,000 each
(3) 50,000 each
(4) 50,000 and 1,000 respectively

65) The given figure shows global biodiversity of proportionate number of species of major taxa of

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vertebrates. Identify A,B, C and D correctly :-

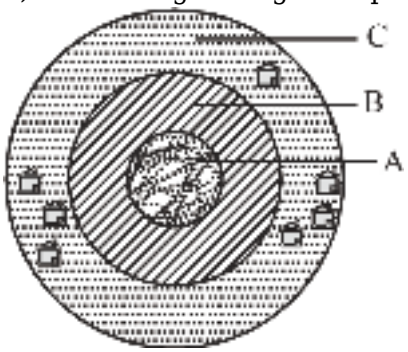
(1) A-Mammals ; B – Birds ;
C – Reptiles ; D – Amphibians

(2) A-Amphibians ; B – Reptiles ;
C – Birds; D – Mammals

(3) A-Mammals ; B – Reptiles ;
C – Birds ; D – Amphibians

(4) A-Fishes ; B – Mammals ;
C – Birds ; D – Reptiles

66) Refer the given figure representing different zones of a biosphere reserve.



Choose the correct answer as the statements given below.
(i) Limited human activity is allowed such as for research and education.
(ii) An active co-operation occurs between reserve management and local people for activities like cropping, settlements etc.
(iii) No human activity is allowed.

	(i)	**(ii)**	**(iii)**
(1)	A	B	C
(2)	B	C	A
(3)	C	A	B
(4)	C	B	A

(1) 1

(2) 2

(3) 3

(4) 4

67) The four (A, B, C and D) given about the indian map, find out the national park/santuaries, and

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conserved animal :-

Location		**National Park/ Sancturies**		**Conserved Animal**		**Location**
A	(a)	Gir	(i)	Royal bengal tiger	P	Gujarat
B	(b)	Jim corbett	(ii)	One horn rhino	Q	Kerala
C	(c)	Periyar	(iii)	Asiatic lion	R	Assam
D	(d)	Kaziranga	(iv)	Elephent	S	Uttarakhand

(1) A-b-iii-P, B-d-i-S, C-a-ii-R, D-c-iv-Q

(2) A-b-i-S, B-d-ii-R, C-a-iii-P, D-c-iv-Q

(3) A-d-i-R, B-a-iii-Q, C-b-iii-P, D-c-ii-S

(4) A-c-ii-P, B-b-iv-Q, C-d-iv-R, D-a-iii-S

68)

(A) 2-Cow 3-Rat 4-Insect

(B) 5-Cow 5-Insect 5-Bird

(C) 5-Cow 5-Insect 5-Bird 5-Snake

(D) 10-Cow 10-Insect 1-Goat 1-Frog

In above diagram four areas are given. In which Area has maximum species diversity

(1) A

(2) C

(3) B

(4) D

69) **Statement-I :** Among animals, insects are the most species rich taxonomic group.
Statement-II : The number of fungi species in the world is more than the combined total of the species of fishes, amphibians, reptiles and mammals.

(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Statement-I & Statement-II both are incorrect.

70) Two honey bees species are competing for the same nectar of a flower in a garden to survive and co-exist together, they may avoid competition in the same garden by :

(1) Predating on each other.

(2) Increasing time spent on attacking each other.

(3) Choosing different foraging patterns.

(4) Feeding at the same time.

71) **Statement-I :** Alien species invasion is the most important cause driving animals and plant to extinction.
Statement-II : Many species extinctions in the last 500 years (Steller's Sea cow, Passenger Pigeon) were due to Habitat loss and fragmentation.

(1) Statement I and II both are correct

(2) Statement I and II both are incorrect

(3) Only Statement I is correct

(4) Only Statement II is correct

72) **Statement-I :** National park, Wild life sanctuaries and biosphere reserves are the example of Ex-situ conservation.
Statement-II : Exploring molecular, genetic and species level diversity for the products of economic importance known as bioprospecting.

(1) Statement I and II both are correct

(2) Statement I and II both are incorrect

(3) Only Statement I is correct

(4) Only Statement II is correct

73) **Statement-I :-** The conventional taxonomic methods are not suitable for identifying microbial species.
Statement-II :- The rate of extinction in sixth episode of extinction is less than the previous episodes of extinction.

(1) Both Statement-I & Statement-II are false

(2) Statement-I is correct but Statement-II is incorrect

(3) Statement-I is incorrect but Statement-II is correct

(4) Both Statement-I & II are correct.

74) Match the column I and coloumn II

	Column-I		**Column-II**
(a)	Dodo	(i)	Africa

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(b)	Quagga	(ii)	Bali, Javan, Caspian
(c)	Thylacine	(iii)	Mauritius
(d)	Steller's Sea Cow	(iv)	Australia
(e)	Tiger sub species	(v)	Russia

(1) (a)- iv (b)- ii (c)- i (d)- v (e) -iii

(2) (a)- iii (b)- iv (c)- ii (d)- i (e) -v

(3) (a)- iii (b)- i (c)- iv (d)- v (e) -ii

(4) (a)- iii (b)- v (c)- iv (d)- i (e) -ii

75) In which of the following both pairs have **correct** combination :-

(1) In situ conservation – Seed Bank
Ex-situ conservation – National park

(2) In-situ conservation – Cryopreservation
Ex-situ conservation – Wild life santuary

(3) In-situ conservation – Biodiversity hot spot
Ex-situ conservation – Wild life safari park

(4) In-situ conservation – Tissue culture
Ex-situ conservation – Sacred groves

76) Match the Column-I and Column-II with correct options -

Column-I		**Column-II**	
(A)	Kyoto Protocol	(i)	Biodiversity Conservation
(B)	Montreal Protocol	(ii)	Climate Change
(C)	The Earth Summit	(iii)	Sustainable development
(D)	World Summit	(iv)	Ozone Protection
		(v)	Wetland Conservation

(1) (A) - (ii); (B) - (iv); (C) - (i); (D) - (iii)

(2) (A) - (iv); (B) - (ii); (C) - (v); (D) - (iii)

(3) (A) - (iv); (B) - (ii); (C) - (v); (D) - (i)

(4) (A) - (ii); (B) - (iv); (C) - (iii); (D) - (i)

77)

Match the following column :

	Column-I		**Column-II**
A.	Rivet-Poper Hypothesis	(i)	Edward Wilson
B.	Biodiversity term	(ii)	Alexander Von Humboldt

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C.	Species-area relationship	(iii)	A.G.Tansley
D.	Ecosystem term	(iv)	Paul Ehlrich

(1) A-iv, B-iii, C-i, D-ii

(2) A-ii, B-iv, C-iii, D-i

(3) A-iv, B-i, C-ii, D-iii

(4) A-iii, B-ii, C-i, D-iv

78) The annual net primary productivity of the whole biosphere is :-

(1) 190 billion tons

(2) 160 billion tons

(3) 170 billion tons

(4) 150 billion tons

79) Which of the following statement(s) regarding energy flow is/are false ?
(I) Detritus food chain begins with dead organic matter
(II) In aquatic ecosystem, detritus food chain is the major conduit for energy flow
(III) In terrestrial ecosystem a larger fraction of energy flows through grazing food chain
(IV) Producers belong to the first trophic level of the food chain

(1) II and III only

(2) III and IV only

(3) I and IV only

(4) I and II only

80) What is the percentage of photosynthetically active radiation (PAR), in the incident solar radiation :-

(1) 100%

(2) 50% or less

(3) 1-5%

(4) 2-10%

81) Which statement is **incorrect** ?

(1) Trophic level represent functional level not a species as such.

(2) A same species may occupy more than one trophic level in same ecosystem at same time.

(3) In most ecosystem, pyramids of number, or energy and biomass are upright.

(4) Energy at higher trophic level is more as compared to lower levels.

82) If you were to count the number of insects feeding on a big tree and number of small birds depending on the insects as also the number of larger birds eating the smaller.
What kind of energy pyramid would you get for food chain :-

(1) 

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(2) 

(3) 

(4) 

83) Ecosystem is an open system for :-

(1) Energy
(2) Minerals
(3) Both 1 and 2 given above
(4) Water

84) What can be correct for following food web :- 

(1) J is primary consumers
(2) C is herbivore
(3) I is primary carnivore
(4) F is carnivore

85) Read statements regarding population interacton:
(A) Biological control methods are adopted in agricultural pest control based on predation
(B) In the rocky intertidal areas of American pacific coast starfish is an important predator.
(C) Some species of frog & insect show camouflagae
(D) Host specific Parasite and host show co-evolution
How many of above statements are **correct** ?

(1) Statement A, B, C, D
(2) Only statement D
(3) Only statement C, D
(4) Only A, B and D

86) Which one of the following is most appropriately defined ?

(1) Amensalism is a relationship in which one species is benefited where as the other is unaffected.
(2) Predator is an organism that catches, kills & eats other organisms for food.
(3) Parasite is an organism which always lives inside the body of other organism and kill it.
(4) Female *Anopheles* mosquito is considered as parasite

87) Which of the following species relation is not an example of detrimental relation ?

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(1) Copepods ↔ Marine fishes

(2) Sparrow ↔ seed

(3) Ticks ↔ Dog

(4) Glomus ↔ Root

88) Identify the correct match from the coloumn I, II and III :-

	Column-I		**Column-II**		**Column-III**
1	Predation	a	Plant pollinator Relation	i	+/–
2	Commensalism	b	Cuckoo & Crow	ii	+0
3	Mutualism	c	Biological control method	iii	+/+
4	Brood Parasitism	d	Sea-anemone Clown fish	iv	+ –

(1) 1-b-i, 2-a-ii, 3-c-iii, 4-d-iv

(2) 4-a-ii, 2-b-iii, 1-c-iv, 3-d-i

(3) 1-c-i, 2-d-ii, 3-a-iii, 4-b-iv

(4) 1-c-i, 2-b-iv, 3-a-iii, 4-d-ii

89) Which one is incorrect :-

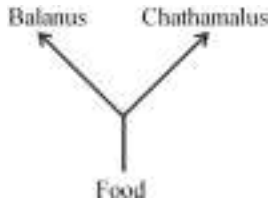
(A)



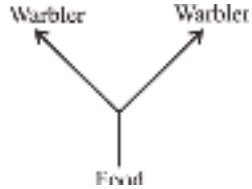
(B)



(C)



(D)



(1) A – Interference competition

(2) B – Showing Gause competitive exclusion principle

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(3) C – Showing intra specific competition

(4) D – Showing resources partitioning

90) Population interactions :-

Organism A	Organism B	Name of interaction
+	+	Mutualism
–	–	A
+	–	Predation
+	–	B
+	0	Commensalism
–	0	C

Find out what could be A, B and C.

(1) A-Amensalism, B-Parasitism, C-Competition

(2) A-Competition, B-Parasitism, C-Amensalism

(3) A-Competition, B-Amensalism, C-Parasitism

(4) A-Amensalism, B-Competition, C-Competition

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ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	3	2	4	2	3	1	2	1	1	3	2	4	1	2	3	4	1	1	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	3	4	2	4	1	3	1	3	1	3	4	2	4	2	2	2	1	3	1
Q.	41	42	43	44	45															
A.	2	4	2	3	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	1	2	3	2	2	2	1	2	2	4	1	4	1	3	3	3	3	4	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	4	4	3	2	1	3	4	3	2	4	2	4	3	4	1	3	2	3	3
Q.	86	87	88	89	90															
A.	3	2	3	2	3															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	4	3	2	3	2	3	2	1	2	3	3	3	4	2	3	2	4	3	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	1	3	1	3	3	2	3	3	2	3	4	2	3	3	2	3	1	3	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	1	1	4	3	2	4	3	3	1	2	4	2	4	4	1	2	1	1	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	1	4	4	3	2	2	2	1	3	2	4	2	3	3	1	3	3	1	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	2	1	3	1	2	4	3	3	2										

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PHYSICS

1) In question it is clearly written that "definitely return". Since scaling is not done, in all the cases we can find atleast a situation in which the particle can not return to its initial position.

2)

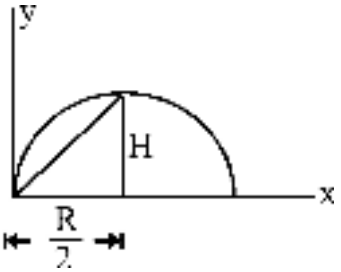
Average velocity = $\frac{\text{Displacement}}{\text{Time}}$

$v_{avg} = \frac{\sqrt{H^2 + \frac{R^2}{4}}}{T/2}$

Here. H = maximum height

$= \frac{v^2 \sin^2\theta}{2g}$

R = range = $\frac{v^2 \sin 2\theta}{g}$



and T = time of flight = $\frac{2v \sin\theta}{g}$

$\Rightarrow \quad v_{avg} = \frac{v}{2}\sqrt{1 + 3\cos^2\theta}$

Alternate solution :

$\langle V_x \rangle = \sqrt{\frac{V\cos\theta + V\cos\theta}{2}} = V \cos\theta$

$\langle V_y \rangle = \sqrt{\frac{0 + V\sin\theta}{2}}$

$\langle V \rangle = \sqrt{V_x^2 + V_y^2} = \sqrt{V^2\cos^2\theta + \frac{V^2\sin^2\theta}{4}}$

$\langle V \rangle = \frac{V}{2}\sqrt{1 + 3\cos^2\theta}$

3)

As $t = x^2 + x \Rightarrow dt/dx = 2x + 1$

$\Rightarrow \quad v = (2x + 1)^{-1} \quad \ldots (i)$

[as dx/dt = v], Now by chain rule

$a = \frac{dv}{dt} = \frac{dv}{dx}\cdot\frac{dx}{dt} = v\frac{dv}{dx}$

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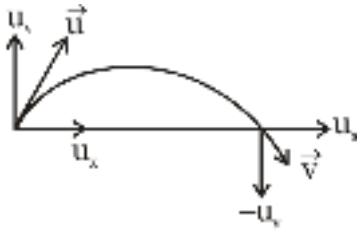
$\Rightarrow \quad a = (2x + 1)^{-1} \frac{d}{dx}(2x + 1)^{-1}$

$= -2\ (2x + 1)^{-3}$

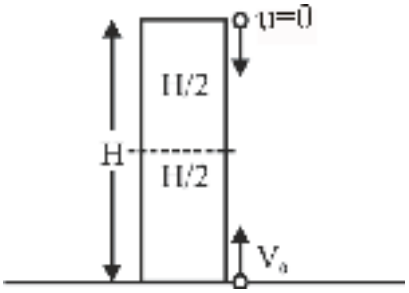
So, Retardation $= -a = 2\ (2x + 1)^{-3}$

4) $\vec{u} = 2i + 3j$

So, $\vec{v} = 2i - 3j$



5)



Time of meeting $t = \frac{H}{V_0 + 0}$

$t = \frac{H}{V_0}$ ...(1)

For ball from ground using eq. (2)

$S = ut + \frac{1}{2}at^2$

$\frac{H}{2} = V_0(t) - \frac{1}{2}g(t^2)$ ...(2)

$\frac{H}{2} = V_0\left(\frac{H}{V_0}\right) - \frac{1}{2}g\left(\frac{H}{V_0}\right)^2$

$V_0^2 = gH \qquad V_0 = \sqrt{gH}$

6)

$v = u + at$

$0 = 15 - 2t$

$t = 7.5$ sec

from 7.5 sec to 8 sec

$S = \frac{1}{2}at^2 = \frac{1}{2} \times 2 \times \frac{1}{4}$

$S = \frac{1}{4}$

Distance travelled in 8th sec. $= 2\left(\frac{1}{4}\right) = 0.5$ m

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7)

Question Asking About:
In this question we have to find the maximum velocity of car for given conditions

Given Data:
Initial velocity of car (starts from rest) m = 0

Acceleration of car = a

Deceleration of car = b

Total time taken = t

Key Concept:
Equation of motion with constant acceleration.

Formula:

V = μt at

Solution/Explanation/Calculation:
Let at acceleration for time t_1 and deceleration for time t_2 then

$t_1 + t_2 = t$ ....(1)

from v = μ + at

v = final velocity

μ = Initial velocity

a = acceleration

t = time

for acceleration

$v_{max} = 0 + at_1$ $(v = v_{max})$

$v_{max} = at_1$ ...(2)

for deceleration :

The velocity at the end of time t_1 i.e v_{max} will be the initial velocity for time internal t_2 and final velocity will be 0.

for acceleration

from v = m + at

$\Rightarrow 0 = v_{max} + (-b / t_2)$

$\Rightarrow 0 = at_1 - bt_2$ [$v_{max} = at_1 \rightarrow$ from – 2, a = -b]

$\Rightarrow t_2 = \frac{at_1}{b}$

Putting value of t_2 in equation (1)

$\Rightarrow t_1 + t_2 \Rightarrow t_1 + \frac{at_1}{b} = t$

$\Rightarrow \frac{bt_1 + at_1}{b} = t \Rightarrow t_1 = \frac{bt}{a+b}$ ....(3)

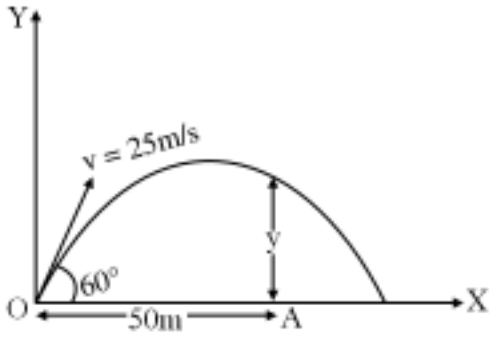
Putting value of t_1 in equation (2)

$V_{max} = at_1 \Rightarrow \frac{abt}{(a+b)} = \frac{abt}{a+b}$

Conclusion:
Hence, option (1) is correct.

8) $y = 12x\left(1 - \frac{x}{16}\right)$
compare with
$y = x\tan\theta\left(1 - \frac{x}{R}\right)$
R = 16 m

9) $S = \frac{1}{2}\left(\frac{\alpha\beta}{\alpha+\beta}\right)t^2$
t = 30 s

10)


$V_x = 25\cos 60° = 12.5$ m/s
$V_y = 25\sin 60° = 12.5\sqrt{3}$ m/s
Time to cover 50 m distance $t = \frac{50}{12.5} = 4\,\text{sec}$
The vertical height y is
$y = V_y t - \frac{1}{2}gt^2 = 12.5\sqrt{3} \times 4 - \frac{1}{2} \times 9.8 \times 16 = 8.2\text{m}$

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11) $v_x = u_x = 147\cos 60^\circ$ and $v_y = u_y - gt = 147\sin 60^\circ - (9.8)t$
If its inclination with the horizontal is 45° then $v_x = u_y$
So, $t = \frac{147\left(\sqrt{3}-1\right)}{2\times 9.8} = 5.49s$

12) $S = \frac{1}{2}gt^2 = \frac{1}{2}\times 10\times 9 = 45m$
$45 = \frac{1}{2}g(2n-1)$

13)

Given initial velocity = ν_0
Final velocity = 0
$\alpha = -x^2$
Now we know that $\alpha = \frac{d\nu}{dt} = \frac{d\nu}{dt}\times\frac{dx}{dx} = \nu\frac{d\nu}{dx} = -x^2$
$\nu d\nu = -x^2 dx$
Integrating we get
$\int_{\nu_0}^{0}\nu d\nu = \int_0^x -x^2 dx$
$\left(\frac{\nu^2}{2}\right)_{\nu_0}^{0} = -\left[\frac{x^3}{3}\right]_0^x$
$-\frac{\nu_0^2}{2} = -\frac{x^3}{3}$
$x = \left(\frac{3\nu_0^2}{2}\right)^{\frac{1}{3}}$

14) Let r be the length of seconds hand.
Distance travelled in 1 sec $= \frac{2\pi r}{60} = \frac{\pi r}{30}$
Now, we want to know the distance travelled in 27 seconds $= 27\times\frac{\pi r}{30}$
$= 27\times\frac{\pi\times 5}{30}(\because r = 5cm)$
$= \frac{9\pi}{2}cm$

15)

$\int_0^v dv = \int_0^t (6t+5)dt$

$\Rightarrow a = 6t + 5$ $\Rightarrow v = 3t^2 + 5t$

$$\int_0^s ds = \int_0^2 (3t^2 + 5t)dt$$

$$S = 3\left[\frac{t^3}{3}\right]_0^2 + 5\left[\frac{t^2}{2}\right]_0^2 = 8 + 10 = 18\,m$$

16) $T = \frac{U_y}{g} \Rightarrow 2 = \frac{U_y}{g}$

$U_y = 20$ m/s

at t = 1 $v_y = 10$ m/s

$v_x = 10$ m/s

$v = \sqrt{(10)^2 + (20)^2} = 10\sqrt{5}m/s$

17) $a = \frac{dV}{dt} = \text{slope}$

Slope : negative ⇒ zero ⇒ Positive

Slope : First decrease them increases

18) $T = \frac{2u\sin\theta}{g}$ 3 + 5

⇒ u sin θ = 40 .....(1)

and u cos θ × 2 = 80

⇒ u cos θ = 40 .....(2)

from equation (1) and (2)

tan θ = 1, θ = 45º

19)



$\vec{S} = \left(-30\hat{j} + 40\hat{i}\right)$ km, $\left|\vec{S}\right| = \sqrt{(30)^2 + (40)^2}$

$\left|\vec{S}\right| = 50km$

$\tan\theta = \frac{30}{40} = \frac{3}{4}$

θ = 37º South of east

20) Displacement = 0

as initial & final position of object is same.

21)

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$y = mx \Rightarrow V^2 = \frac{4}{4}t \Rightarrow V^2 = t \Rightarrow V = \sqrt{t}$

$a = \frac{dV}{dt} = \frac{1}{2\sqrt{t}} = \frac{1}{2\sqrt{4}} = \frac{1}{4}\ m/s^2$

22) $\left(\vec{V}_i\right) = (u\cos\theta)\hat{i} + (u\sin\theta)\hat{j}$

$\left(\vec{V}_f\right) = (u\cos\theta)\,\hat{i} - (u\sin\theta)\hat{j}$

$\left(\vec{V}_{top}\right) = (u\cos\theta)\,\hat{i}$

$|\vec{V}_i|$ = Initial speed = u

$|\vec{V}_f|$ = final speed = u

$|\vec{V}_{top}|$ = speed at top = u cos θ (A)

(B) $\overrightarrow{\Delta V} = \vec{V}_f - \vec{V}_i$ & $\Delta|\vec{V}| = |\vec{V}_f| - |\vec{V}_i| = 0$

$|\overrightarrow{\Delta V}| = (2u\sin\theta)\hat{j}$

$|\overrightarrow{\Delta V}| = 2u\sin\theta$

(C) $|\vec{V}_i| - |\vec{V}_{top}| = u - u\cos\theta$

(D) $|\vec{V}_i - \vec{V}_{top}| = u\sin\theta$

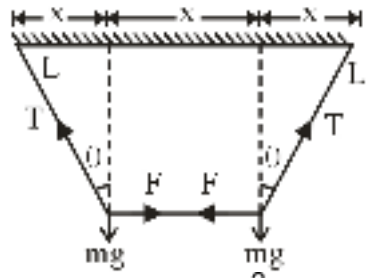
23) $v = \frac{dx}{dt} = 2t^2 - 4t + 2 = 2(t-1)^2$ which is always positive so direchon of velocity remains unchanged.

24)

Total charge = $\int_0^R \rho dv = \int_0^R \frac{\rho_0 r}{R}(4\pi r^2)dr = \pi R^3 \rho_0$

25) To cover the charge we need 12 surface, $\phi = \frac{q}{12\epsilon_0}$

26) $\tan\theta = \frac{F_E}{mg}, T\sin\theta = F$



$\Rightarrow T\left(\frac{x}{L}\right) = \frac{1}{4\pi\varepsilon_0}\frac{Q^2}{x^2}$

$$T = \frac{1}{4\pi\varepsilon_0}\frac{Q^2L}{x^3}$$

27)

Total charge $= \int \lambda dx$

$$= \int_0^L \frac{\lambda_0 x}{2L} dx = \frac{\lambda_0}{2L}\left[\frac{x^2}{2}\right]_0^L$$

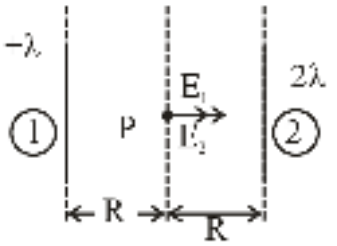
$$= \frac{\lambda_0 L}{4}$$

28)

At mid point P

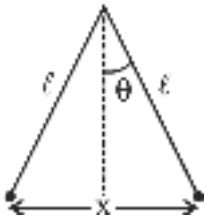
$$E = E_1 + E_2 = \frac{\lambda}{2\pi\lambda_0 R} + \frac{2\lambda}{2\pi\varepsilon_0 R}$$

$$E = \frac{3\lambda}{2\pi\varepsilon_0 R}$$



29) for small angle

$$\tan\theta \approx \sin\theta = \frac{Fe}{mg}$$



$$\frac{x}{2\ell} = \frac{Kq^2}{x^2(mg)}$$

so, $q^2 \propto x^3$ or $q \propto x^{3/2}$

30) $$\phi = \frac{\lambda \ell_{enclosed}}{\in_0}$$

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$$= \frac{8.85 \times 10^{-6} \times 5}{8.85 \times 10^{-12}}$$
$$= 5 \times 10^{6} \text{ V-m}$$

31) **Concept**
The electric field due to a away from the charge; while the electric field due to a toward the charge. We are looking for the combination of charges where the field is positive along the x-axis
Formula
The direction of the field depends on the sign of the charge.
• For a positive charge, the electric field points away from the charge.
• For a negative charge, the electric field points towards the charge.
Calculation
To ensure the electric field is negative at a point between Q_1 and Q_2, the direction of the fields due to both charges must add up in the negative direction of x. This can happen if Q_1 is negative and Q_2 is positive and the graph is symmetric so $|Q_1| = |Q_2|$
Answer: Option 3 (Q_1 is negative and Q_2 is positive with $|Q_1| = IQ_2|$

32)

• (A) Like charges – (ii) Repel each other

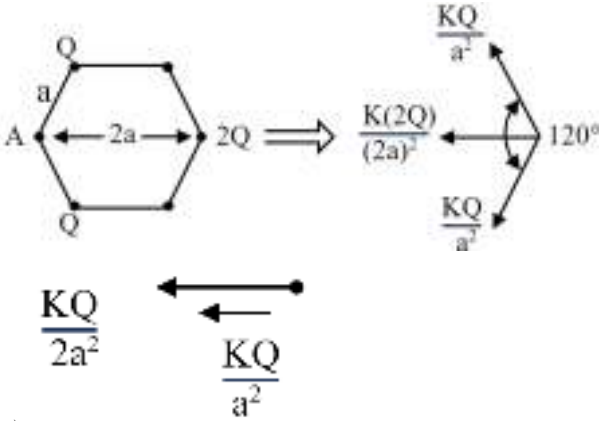
• (B) Unlike charges – (iii) Attract each other

• (C) Charge can be – (iv) Neutralised if they are equal and opposite

• (D) Charge are – (i) of two types.

Hence, option (4) is correct.

33) **Electric Field Calculation**
The electric field at point A is the vector sum of the electric fields due to all charges at the vertices of the hexagon.



⇒
$$E_{net} \Rightarrow \frac{kQ}{2a^2} + \frac{kQ}{a^2}$$
$$\Rightarrow \frac{3}{2}\frac{kQ}{a^2}$$
Calculation of Electric Potential:

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The potential at point A is the sum of the potentials due to each charge.

• For charge Q at distance a:

$$V_Q = \frac{kQ}{a}$$

• For charge 2Q at distance 2a:

$$V_{2Q} = \frac{k(2Q)}{2a} = \frac{kQ}{a}$$

Total potential at point A:

$$V_A = \frac{kQ}{a} + \frac{kQ}{a} + \frac{kQ}{a}$$

$$\Rightarrow V_A = \frac{3kQ}{a}$$

34)

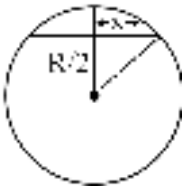
For vacuum, $F = \frac{Kq_1q_2}{r^2}$ ..(1)

For partial filling of dielectric, $r_{eff} = \left(r - \frac{r}{2}\right) + \frac{r}{2}\sqrt{4}$

$$r_{eff} = \frac{3r}{2}$$

$$F' = \frac{Kq_1q_2}{\left(\frac{3r}{2}\right)^2} = \frac{4F}{9}$$

35)



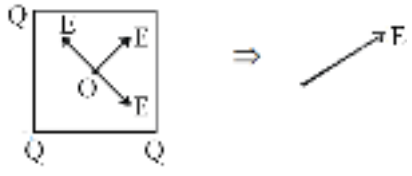
$$\phi = \frac{q_{in}}{\in_0}$$

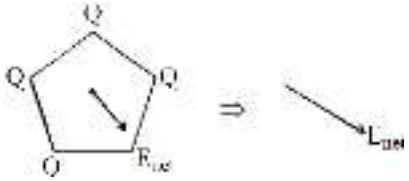
$q_{in} = \lambda \cdot 2x$

$$2x = 2\sqrt{R^2 - \frac{R^2}{4}} = \sqrt{3}R$$

So, $\phi = \frac{\sqrt{3}\lambda.R}{\in_0}$

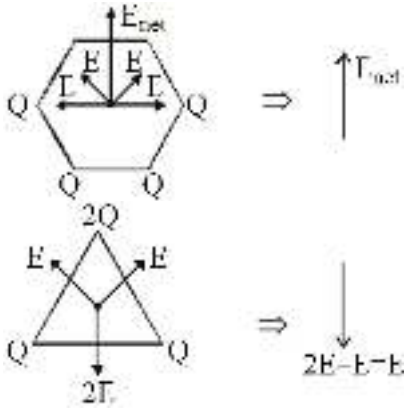
36)

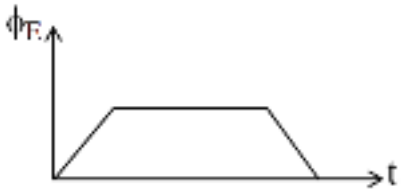




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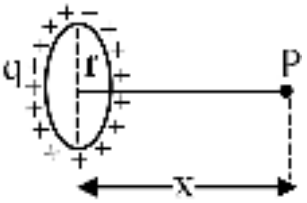
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37) 

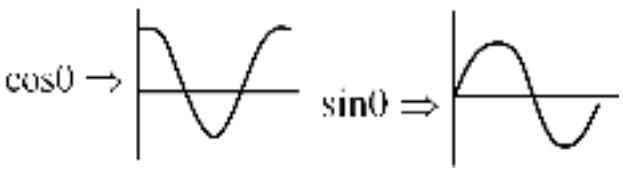
38) $\phi = \frac{q}{\epsilon_0} \frac{2.15 \times 10^{-6}}{8.85 \times 10^{-12}} = \frac{10^6}{4.1}$
$\approx 2.5 \times 10^5$

39) $\frac{F_1}{F_2} = \frac{(\theta_1.\theta_2)}{\left(\frac{\theta_1+\theta_2}{2}\right)^2} \Rightarrow \frac{F_1}{F_2} = \frac{4\theta_2}{\theta_1}$

40) 

$E_P = \frac{kqx}{(r^2 + x^2)^{\frac{3}{2}}}$

41) Electric field lines are equidistant and parallel. Electric field is uniform.

42) 

43) $x^2 + y^2 = R^2$ in x-y plane
$\pi R^2 \hat{k} = 4\pi \hat{k}$
(R = 2 m)

$d = \vec{E}.\vec{A} = 4\gamma\pi$

44)

$R \Rightarrow \infty$

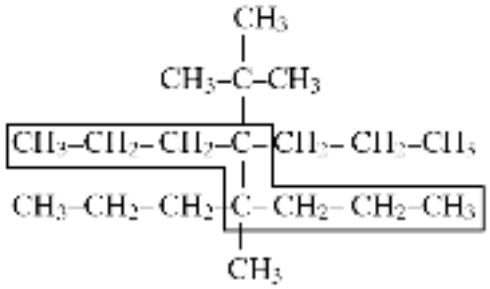
$\phi = \frac{q}{2\epsilon_0}$

3rd option correct.

45) $E = \frac{kq}{r^2}$

3rd option correct.

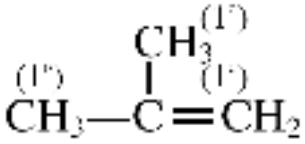
CHEMISTRY



46)

47)

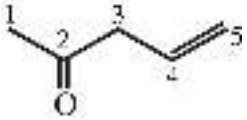
Nu. Starts from OH group.



48)

49)

Priority sequence of functional groups.



50)

51)

COOH is PFG. one CHO is included in PCC.

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52)

$CONH_2$ is PFG

53)

– OH is functional group

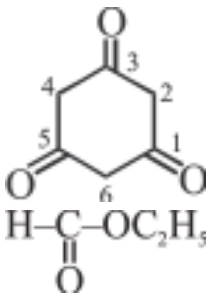
54)

Open chain part is PCC.

55)

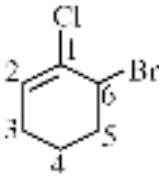
Nu starts from OH group.

56)

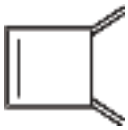


Cyclohexane-1,3,5-trione

Ethyl methanoate



6-Bromo-1-chloro cyclohexene

57)  All C-atom are sp^2

58)

In p-block all type's of element's are present.

59)

Amphoteric oxide : CrO_3, V_2O_5, MnO_2

60)

1) In a group M.C decrease

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2) Electropositive $\alpha \frac{1}{IE}$
3) EN > Z_{eff} (does not depend on stability)
4) $IP_1 < IP_2 < IP_3 < ...$

61)

IE $O^- < S^{\ominus}$
$(EA)_0 = (IE)_{0^{\ominus}} < (EA)_4 = (IE)_5$

62)

In Cl-atom additional e^- get less repulsion.

63)

EA : O < F
EA : O < N (dneto soluble $2s^2\ 2p^3$ configuration of N)

64)

CO_3^{2-} follows octet rule ⇒ neither hypo valent nor hyper valent.

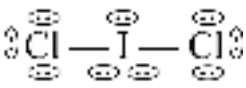
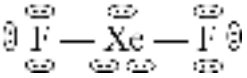
65)

CCl_6^{-2} does not exist.

66) Iodine's ground state has only one unpaired electron. To form IF_3, iodine needs three unpaired electrons. which occurs in the first excited state.
Final Answer - Option (1)

67)

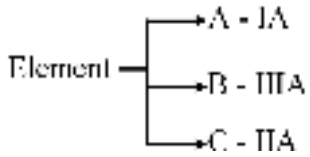
Since all the individual statements are correct, this option is also correct.

68) $I_3^{\ominus}$: 
$ICl_2^{\ominus}$: 
XeF_2 : 

69)

(1) px + px along X-axis: This results in head-on overlap, forming a σ bond, not a π bond.

(2) px + dxz along X-axis: The px orbital is along the X-axis. The dxz orbital has lobes in the XZ plane. Sideways overlap is not possible, forming a σ bond.
(3) py + dxy along X-axis: The py orbital is along the Y-axis. The dxy orbital has lobes between the X and Y axes. Sideways overlap is possible, forming a π bond.
(4) s + pz along X-axis: The s orbital is spherical. The pz orbital is along the Z-axis. There will be no effective sideways overlap to form a π bond along the X-axis. forming a σ bond, not a π bond.
Answer option 3

70) 

71)

$Al(OH)_3$ - Amphoteric
$Fe(OH)_3$ - Basic

72)

Al = 1.5, Ga = 1.6, C = 2.5, N = 3.0
Br = 28

73)

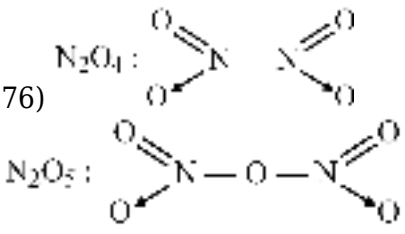
$Cu(OH)_2$ – Basic nature

74)

PbO_2 – Amphoteric
SO_2, Mn_2O_7, Cl_2O_7 – Acidic

75)

Valence with respect to hydrogen
1 2 3 4 3 2 1

76) 

77) $P_z + P_z \xrightarrow{Z\text{–axis}} \sigma$

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$P_z + d_{yz} \xrightarrow{Z\text{–axis}} \otimes$
$P_y + d_{yz} \xrightarrow{Z\text{–axis}} \lambda$

78)

N ≡ N, –C ≡ C –/ >C = C<

79) $P_x + P_x \xrightarrow{Z\text{–axis}} \lambda$
$P_x + d_{xz} \xrightarrow{y\text{–axis}} \lambda$

80) $IP : F^{\ominus} > S^{\ominus} > O^{\ominus}$
$IP : O^{-} < Br^{\ominus}$

81)

3 A(g) → 2B(g) + 3C(g) + D(g)
P_0
$P^0 - x \qquad \frac{2}{3}x \qquad x \qquad \frac{x}{3}$
$P_t \Rightarrow P^0 - x + \frac{2}{3}x + x + \frac{x}{3} = P^0 + x$
$x = (P_t - P_0)$
$K = \frac{1}{t} \ln \frac{P_0}{P_0 - (P_t - P_0)}$
$K = \frac{1}{t} \ln \frac{P_0}{2P_0 - P_t}$

82) $\boxed{1,2} \frac{1}{0.5} = (1)^x \left(\frac{2.4}{1.2}\right)^y (1)^z \therefore y = 1$
$\boxed{2,3} \frac{2}{1} = (2)^x (1)^1 (1)^z \therefore x = 1$
$\boxed{1,4} \frac{4.5}{0.5} = (1)^1 (1)^1 \left(\frac{0.9}{0.3}\right)^z \therefore z = 2$
overs all order = 4

83)

$\rightarrow t_{t/2} \propto \frac{1}{(A_0)^{n-1}}$
$\rightarrow A_t = A_0 e^{-kt}$ first order

→ Pseudo unimolecular molecularity = 2 order = 1

84)

x → 3y

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t = 0 a 0
t_{eq} a – x 3x
a – x = 3x
a = 4x
$x = \frac{a}{4}$

85)

r = K(A)'
$1.5 \times 10^{-2} = K \times 0.5$
$K = 3 \times 10^{-2}$
$t_{1/2} = \frac{0.693}{3 \times 10^{-2}} = 23.1 \text{ min}$

86) (a) Incorrect
(b) T ↑ effective collision ↑ ROR ↑ correct
(c) correct
(d) correct

87)

$10^{12}\, e^{-4000/T} = 10^{13} \frac{-5500}{T}$
$e \frac{+5500-4000}{T} = 10$
$\frac{1500}{T} = 2.303 \log 10$
$T = \frac{1500}{2.303} = 651.32 \text{ k}$

88) a, c are correct, b, d incorrect.
(a) $K = Ae^{-E_a/RT}$ E_a↑ K↓
(b) $\log\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$

89) $K = \left(\frac{K_1}{K_2}\right)^{3/2} (K_3)^2$
$A_{overall} = \left(\frac{A_1}{A_2}\right)^{3/2} (A_3)^2 = \left(\frac{10^6}{10^2}\right)^{3/2} (10^4)^2 = 10^{14}$
$E_a = \frac{3}{2} \times 40 - \frac{3}{2} \times 60 + 2 \times 30 = 30KJ$

90) (a) $k = Ae^{-Ea/Rt}$ when $E_a = 0$ ⇒ k = A constant
(b) T = ∞ ⇒ K = A = 1.5×0^{14}

(c) $\log k = \log A - \frac{Ea}{2.303R}$

So $\frac{Ea}{2.303 \times 2} = 2000 \Rightarrow 9.212$ kcal

BIOLOGY

91) Its an ancestral charecter

92) Ampullary-isthmic junction is typical fertilization site.

93) Oxytocin stimulates uterine and breast contraction.

94) It prevents sperm from reaching urethra.

95) It connecting link

96) LH surge causes ovulation.

97) Acrosome has hydrolytic enzymes.

98) Secreted by trophoblast cells.

99) Sequential development post-fertilization.

100) Initiated by FSH and LH after puberty.

101) Fallopian tubes are cut and tied.

102) Also helps gain motility.

103) No hematopoiesis in placenta.

104) It is a vestigial organ

105) Closely related groups evolve similarly in different region.

106)

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NCERT (XII) ; Page No. # 31
Helps in spermiogenesis

107)

NCERT (XII) ; Page No. # 38
Prolactin promote lactogenesis

108)

Testis from gametes

109)

9 months $\pm$ 10 days

110)

Mucosa of uterus

111)

Responsible for estivation of collagenase

112) One ancestor given many new species indifferent habitats is adaptive radiation.

113)

Made of Meternal of and Foetal Tissue

114)

Function of structures

115)

Its an IUCD

116)

Contains progesterone

117)

Uterus is female reproductive organ.

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118)

These are sex hormone

119) Protein & DNA show similar sequence pattern.

120)

NCERT (XII) ; Page No. # 111, 112
Life originates from inorganic substance

121)

Prevent fusion of gamete

122)

Neutralise H^+

123)

Interstitial cell of testis

124)

Breakdown of CL

125)

Days after fertilization

126)

Breakdown of CL

127)

Epididymis

128)

NCERT (XII) ; Page No. # 113
Pinna & Nails are not vestigeal organ.

129)

Anti estrogen drug prevent implantation

130)

LH surge promote ovulation

131)

Fructose is source of energy to sperm.

132)

Placenta release hormones in blood.

133)

Oparin Haldane theory

134)

Match method of contraception

135)

Spermatogenesis is sperm formation.

136)

S = species richness, C = Y intercept, Z = regression coefficient, A = area
The Correct answer is: 2

137)

Question Explanation :
The question is asking to identify which of the given options is a hot spot of Biodiversity in India.

Concept :
Question is based on in situ conservation.

Solution :
The correct option is (4) ; Western Ghats.

Final Answer :
Option (4)

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138)

Module - 8, Pg. No. # 111

139)

Explaining the Question :-
The question is asking which pair of geographical areas in India exhibits the **maximum biodiversity**. To answer this, we need to look at regions that are known for their diverse ecosystems, endemic species, and varied flora and fauna.
Concept:-
Solution :-
This is the **correct answer.** Both the **Eastern Himalayas** and **Western Ghats** are biodiversity hotspots. The **Eastern Himalayas** are rich in flora and fauna, with many endemic species and unique ecosystems. The **Western Ghats** are a UNESCO World Heritage site and are known for their rich biodiversity, including many endemic species of plants, animals, and insects.

140)

NCERT-XII, Pg. No. # 224

141)

NCERT-XII, Pg. No. # 225
Question Explanation :
This question is asking about All of the following are included in 'Ex-situ conservation' except.
Solution :
Secred groves are an example of in-situ conservation because they are areas where biodiversity is conserved within its natural habitat, often due to cultural or religious significance.
The other option (Wildlife safari parks, Botanical gardens and seed banks) are all examples of ex-situ conservation, where species are conserved outside their natural habitats.
Final Answer : Option (2)

142)

NCERT-XII, Pg. No. # 225

143)

NCERT-XII, Pg. No. # 217

144)

Module-8, Pg. No. # 111

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145)

Module - 8, Pg. No. # 108

146)

Explaining the Question :-
The question asks which of the statements regarding biodiversity and ecological characteristics is true. You need to evaluate each option for its accuracy.
Concept:-
Solution :-
1. 32% of amphibian species face the threat of extinction.
This statement is true. According to the International Union for Conservation of Nature (IUCN), about **32% of amphibian species** are currently facing the threat of extinction due to factors like habitat loss, climate change, and diseases

147) NCERT XII, Pg. # 262, Para 15.1.2

148) NCERT XII Pg.# 264

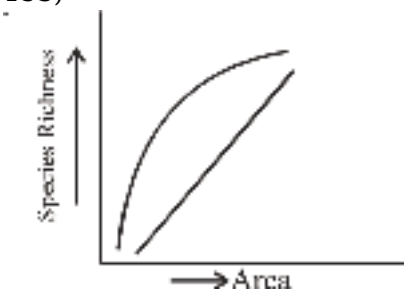
149) NCERT XII, Pg. # 224, 222

150) NCERT-Page No. 261

151) NCERT-Page No. 265

152) NCERT Pg. # 261, 287 (4)

153)



The correct answer is : 4

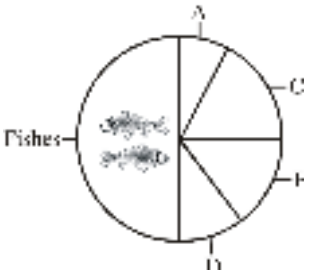


154)

Explaining the Question :-
This question asks for the number of genetically different strains of rice and varieties of mango found in India. It highlights the genetic diversity present in these two important agricultural crops in the country.
Concept:-
Solution :-
• India is home to about 50,000 genetically distinct strains of rice and around 1,000 varieties of mango. These figures reflect the rich biodiversity and agricultural heritage of the country, showcasing the diverse cultivation practices and adaptations in different regions.

155)

Solution :



A-Mammals ; B – Reptiles ;
C – Birds ; D – Amphibians
The correct Answer -3

156)

NCERT-XII, Pg. No. # 224

157)

Module - 8, Pg. No. # 111

158)

Module - 8, Pg. No. # 74

159)

Explain Question : asking about two statements
Concept : This question is based on Number of Species On Earth And Pie Chart
Explanation : Correct Answer: 1. Statement-I & Statement-II both are correct.
Explanation:
Statement-I: Correct
Insects are the most species-rich animal group, making up more than half of all described animal species.
Statement-II: Correct
The estimated number of fungal species (~1.5 million) exceeds the combined total of fishes,

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amphibians, reptiles, and mammals.
Final Answer: 1 Correct Answer

160) NCERT-XII, Pg. # 235

161) NCERT XII, Pg. # 223

162) NCERT Pg # 267

163)

NCERT - XII, Pg.# 260, 264

164)

Explain Question : asking about disappearance of species
Concept : This question is based on Loss Of Biodiversity And Jhum Cultivation
Explanation : In the context of biodiversity and extinction reports, it's often stated that 37 species have disappeared in the last twenty years.
Correct Answer : 4. 37 species.

165) NCERT - Pg. # 224, 225

166)

NCERT-XII, Pg. No. # 225

167)

NCERT-XII, Pg. No. # 220

168)

NCERT page no. 207

169)

NCERT-XII, Pg. No. # 210

170)

NCERT-XII, Pg. No. # 209

171)

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NCERT-XII, Pg. No. # 209

172)

NCERT-XII, Pg # 213

173)

Module - 8, Pg. No. # 107

174)

NCERT-XII, Pg. No. # 210

175)

NCERT-XII, Pg. No. # 197

176)

NCERT-XII, Pg # 233,235

177)

NCERT-XII, Pg. No. # 201

178)

NCERT-XII, Pg. No. # 197

179)

NCERT-XII, Pg. No. # 198

180)

NCERT-XII, Pg. No. # 197

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